

BIO 201 · Plant Physiology: Photosynthesis (Course Notes)

Photosynthesis: an overview

Photosynthesis is the process by which green plants, algae and some bacteria convert light energy into chemical energy stored in glucose. It takes place in chloroplasts, organelles containing the pigment chlorophyll, which absorbs mostly red and blue light and reflects green. The overall balanced equation is $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. Photosynthesis occurs in two connected stages: the light-dependent reactions in the thylakoid membranes, and the Calvin cycle in the stroma.

Light-dependent reactions

The light-dependent reactions occur in the thylakoid membranes. Photosystem II absorbs photons, exciting electrons that travel down the electron transport chain toward photosystem I. Water is split in a process called photolysis, releasing oxygen as a by-product and supplying replacement electrons. The energy released as electrons move down the chain pumps protons into the thylakoid lumen, creating a proton gradient. ATP synthase uses this gradient to produce ATP in a process called chemiosmosis, and NADP^+ is reduced to NADPH.

The Calvin cycle

The Calvin cycle, also called the light-independent stage, takes place in the stroma of the chloroplast. It uses the ATP and NADPH produced in the light-dependent reactions to fix carbon dioxide into organic molecules. The enzyme RuBisCO catalyses the reaction of CO_2 with ribulose biphosphate (RuBP), a five-carbon sugar, producing two molecules of glycerate 3-phosphate (GP). GP is then reduced to triose phosphate (TP) using NADPH and ATP. Some TP is used to regenerate RuBP; the rest is used to synthesise glucose and other carbohydrates.

Limiting factors

The rate of photosynthesis is limited by light intensity, carbon dioxide concentration and temperature. Light intensity increases the rate until saturation, when another factor becomes limiting. CO_2 concentration is often the limiting factor in bright daylight. Temperature affects the rate because the Calvin cycle is catalysed by enzymes; above roughly 40 degrees Celsius, enzymes begin to denature and the rate falls sharply. Farmers exploit these factors in greenhouses, raising CO_2 levels and controlling temperature to increase crop yields.

C4 and CAM adaptations

Some plants have evolved adaptations to hot, dry climates. C4 plants such as maize separate carbon fixation from the Calvin cycle in space: CO_2 is first fixed into a four-carbon compound in mesophyll cells, then transported to bundle-sheath cells where the Calvin cycle runs at high CO_2 concentration. This reduces wasteful photorespiration. CAM plants such as cacti separate the two stages in time: stomata open at night to fix CO_2 into organic acids, and the Calvin cycle runs during the day with stomata closed, minimising water loss.